

NIZHNY TAGIL, Russian Federation

WMO: 282400

Lat: 57.883N

Lon: 60.067E

Elev: 260

StdP: 98.24

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.3	-28.6	-34.3	0.2	-31.2	-31.5	0.2	-28.4	7.9	-15.7	7.2	-13.2	1.6	20	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	28.5	19.6	26.7	18.7	25.0	17.7	20.8	26.6	19.8	25.2	18.7	23.6	2.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	14.0	23.5	17.7	13.1	22.3	16.7	12.3	21.4	61.2	26.6	57.6	25.2	54.1	23.6	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
7.5	6.5	5.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-14.7	-13.0	-5.2	3.0	10.2	14.9	17.1	15.0	9.2	2.7	-6.5	-11.4
	DBStd	12.54	7.30	7.43	5.46	5.03	5.12	4.34	3.84	4.08	4.44	4.90	7.72	7.42
	HDD10.0	3636	767	645	472	216	63	9	2	6	68	230	494	663
	HDD18.3	6081	1025	878	731	459	255	118	70	118	276	484	744	921
	CDD10.0	662	0	0	0	6	69	156	222	161	43	4	0	0
	CDD18.3	66	0	0	0	0	2	16	31	16	2	0	0	0
	CDH23.3	750	0	0	0	1	58	189	332	157	12	0	0	0
	CDH26.7	150	0	0	0	0	3	38	67	41	1	0	0	0
	WSAvg	2.5	2.4	2.6	2.9	2.8	2.5	2.3	1.9	2.1	2.2	2.9	2.9	2.6
Precipitation	PrecAvg	557	23	17	26	30	47	92	80	77	53	40	34	25
	PrecMax	818	44	35	58	88	116	357	175	142	115	84	59	42
	PrecMin	411	3	5	3	2	14	20	12	16	9	13	9	8
	PrecStd	96	11	9	16	24	25	67	37	29	26	19	15	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	2.2	9.7	21.7	26.9	30.0	30.4	30.3	25.0	16.9	8.2	1.9
		MCWB	-0.2	0.3	4.2	12.5	15.7	19.9	20.4	20.2	18.2	9.9	6.2	0.6
	2%	DB	-1.0	0.6	6.3	17.1	25.0	27.5	28.6	27.4	21.1	14.1	5.8	1.1
		MCWB	-2.0	-0.9	2.2	10.2	15.2	18.7	20.0	19.5	14.8	9.2	4.4	-0.1
	5%	DB	-2.3	-1.0	3.8	14.2	22.7	25.7	27.0	24.6	17.9	12.0	3.7	-0.2
		MCWB	-3.3	-2.5	0.6	8.4	14.0	18.1	19.4	18.1	13.7	8.1	2.5	-1.2
	10%	DB	-4.3	-3.1	2.0	11.3	20.0	23.4	25.3	22.4	15.6	9.9	1.7	-1.6
		MCWB	-5.2	-4.5	-0.1	6.2	12.7	16.8	18.6	17.2	12.4	7.1	0.6	-2.7
	0.4%	WB	-0.1	0.7	4.8	12.7	17.6	21.9	22.1	21.3	18.5	11.0	6.6	1.0
		MCDB	1.1	2.0	8.8	19.8	23.7	28.1	27.2	27.8	24.1	14.9	8.1	1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-1.8	-0.7	2.6	10.7	16.1	20.1	21.1	20.1	16.0	9.8	4.6	-0.1
		MCDB	-0.9	0.5	5.7	16.3	23.3	25.7	26.7	26.2	19.3	13.4	5.8	0.8
	5%	WB	-3.2	-2.5	1.1	8.7	14.7	18.6	20.1	18.9	14.3	8.2	2.7	-1.2
		MCDB	-2.3	-1.1	3.0	13.4	21.1	24.0	25.6	23.5	17.1	11.6	3.7	-0.1
	10%	WB	-5.2	-4.4	0.1	6.8	13.4	17.3	19.1	17.7	12.8	7.0	0.7	-2.6
		MCDB	-4.3	-3.2	1.9	10.6	19.2	22.5	24.2	21.3	15.0	9.6	1.5	-1.6
Mean Daily Temperature Range	5% DB	MDBR	6.7	8.2	8.9	9.5	12.0	10.6	11.1	9.9	8.0	5.8	5.3	5.8
		MCDBR	5.9	6.3	10.1	14.0	15.9	14.5	14.4	14.0	11.1	9.9	5.6	4.3
		MCWBR	5.4	5.4	6.9	8.3	8.1	7.1	6.9	6.8	6.3	5.8	4.7	4.0
	5% WB	MCDBR	5.8	5.7	8.0	13.4	14.2	13.3	13.0	12.6	9.8	8.7	5.1	4.6
		MCWBR	5.6	5.0	6.1	8.5	8.3	7.4	6.8	6.5	6.0	5.6	4.6	4.4
Clear-Sky Solar Irradiance	taub		0.280	0.310	0.367	0.406	0.406	0.401	0.427	0.439	0.380	0.346	0.303	0.276
	taud		2.032	2.044	1.955	2.073	2.243	2.322	2.245	2.177	2.310	2.260	2.115	1.990
	Ebn at Noon		529	694	748	792	826	837	802	756	751	658	499	401
	Edh at Noon		57	90	131	138	125	117	124	124	93	74	53	44
All-Sky Solar Radiation	RadAvg		0.56	1.45	2.72	3.94	4.96	5.32	5.28	3.95	2.39	1.14	0.51	0.32
	RadStd		0.07	0.20	0.26	0.50	0.49	0.47	0.50	0.43	0.23	0.19	0.08	0.05

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page